Qatar Diabetes Guideline 2023

Simulated Clinical Guideline — Portfolio / Educational Use Only

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Section 1. Diagnostic Criteria

Type 2 Diabetes Mellitus (T2DM) is diagnosed when fasting plasma glucose is ≥ 126 mg/dL (7.0 mmol/L), 2-hour plasma glucose during a 75g OGTT is ≥ 200 mg/dL (11.1 mmol/L), or HbA1c is ≥ 6.5% (48 mmol/mol), confirmed on repeat testing unless the patient has unequivocal hyperglycemia symptoms.

Screening is recommended for adults in Qatar aged 40 and above, and earlier for individuals with obesity, a first-degree relative with T2DM, or a history of gestational diabetes.

Section 2. First-line Therapy

Metformin is the preferred first-line pharmacologic agent for all patients with Type 2 Diabetes unless contraindicated. Therapy should be initiated alongside structured lifestyle and dietary counselling.

Metformin should be titrated gradually (e.g. starting at 500mg once or twice daily) to reduce gastrointestinal side effects, with a usual maintenance dose of 1500-2000mg per day in divided doses.

Section 3. Metformin Contraindications

Metformin is contraindicated in patients with an eGFR below 30 mL/min/1.73m^2, severe hepatic impairment, or unstable/decompensated heart failure. Metformin should also be discontinued or avoided when gastrointestinal intolerance prevents the patient from tolerating an effective dose.

In patients with an eGFR between 30 and 45 mL/min/1.73m^2, Metformin dose should be reduced and renal function monitored every 3-6 months.

Section 4. Second-line Therapy

When Metformin is contraindicated or not tolerated, second-line therapy should be individualized based on comorbidities:

(a) SGLT2 Inhibitors (e.g. Empagliflozin, Dapagliflozin) are recommended particularly for patients with established cardiovascular disease, heart failure, or chronic kidney disease, given demonstrated cardiorenal benefit.

(b) GLP-1 Receptor Agonists (e.g. Liraglutide, Semaglutide) are recommended particularly for patients with established atherosclerotic cardiovascular disease or where weight reduction is a clinical priority.

Section 5. Target HbA1c and Monitoring

Target HbA1c is generally below 7.0% (53 mmol/mol) for most non-pregnant adults. A less stringent target (e.g. < 8.0%) may be appropriate for patients with limited life expectancy, extensive comorbidity, or a history of severe hypoglycemia.

HbA1c should be reassessed every 3 months until stable at target, then at least every 6 months thereafter.

Section 5.1 Heart Failure Caution

In patients with Type 2 Diabetes and concurrent Heart Failure, Thiazolidinediones (e.g. Pioglitazone) are contraindicated due to the risk of fluid retention and exacerbation of Heart Failure symptoms.

SGLT2 Inhibitors are preferred in this population: large outcome trials have shown a reduction in Heart Failure hospitalization independent of glycemic benefit, and they are considered appropriate even in patients with reduced ejection fraction.

Section 6. Renal Impairment Considerations

SGLT2 Inhibitors may be initiated down to an eGFR of 20 mL/min/1.73m^2 for patients with established chronic kidney disease, given their renal-protective effect, but glycemic efficacy diminishes at lower eGFR.

GLP-1 Receptor Agonists generally require no dose adjustment based on renal function and may be preferred when SGLT2 Inhibitors are not tolerated in patients with advanced chronic kidney disease.